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Development of physics-consistent conditional diffusion model to overcome data scarcity in critical heat flux

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HIGHLIGHTS

- A diffusion model was developed to generate synthetic critical heat flux data for nuclear energy.
- A conditional diffusion model was developed for targeted generation at user-specified conditions.
- The synthetic data was found to be physically consistent with existing mathematical models.

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ABSTRACT

Deep generative modeling provides a powerful pathway to overcome data scarcity in energy-related applications where experimental data are often limited. By learning the underlying probability distribution of the training dataset, deep generative models, such as the diffusion model, can generate high-fidelity synthetic samples that statistically resemble the training data. Such synthetic data generation can significantly enrich the size and diversity of the available training data, and more importantly, improve the robustness of downstream machine learning models in predictive tasks. The objective of this paper is to investigate the effectiveness of diffusion models for overcoming data scarcity in nuclear energy applications. By leveraging a public dataset on critical heat flux which covers a wide range of commercial nuclear reactor operational conditions, we developed a diffusion model that can generate an arbitrary amount of synthetic samples. Since a vanilla diffusion model can only generate samples randomly, we also developed a conditional diffusion model capable of generating targeted critical heat flux data under user-specified thermal-hydraulic conditions. The performance of the diffusion model was evaluated based on its ability to capture empirical feature distributions and pair-wise correlations, as well as to maintain physical consistency. The results showed that both the diffusion model and conditional diffusion model can successfully generate realistic and physics-consistent critical heat flux data. Furthermore, uncertainty quantification results demonstrate that the conditional diffusion model is highly effective in augmenting critical heat flux data while maintaining acceptable levels of uncertainty.

1. Introduction

Deep generative modeling [1,2] provides a powerful pathway to overcome data scarcity in energy-related applications where experimental data are often limited, costly, or difficult to obtain. By learning the underlying probabilistic distributions of the training dataset, deep generative models (DGMs), such as generative adversarial networks (GANs), variational autoencoders (VAEs), and diffusion models (DM), can generate high-fidelity synthetic datasets that statistically resemble the training data, thereby augmenting scarce experimental datasets. Synthetic data generation can significantly increase the size and diversity of the available training data, and more importantly, enhance the robustness of downstream machine learning (ML) models for predictive tasks.

The field of generative modeling was dominated by GANs between 2014 and the early 2020s. The era of GAN dominance began immediately after their introduction in a 2014 paper by Ian Goodfellow and his colleagues, which was later published in "Communications of the ACM" in 2020 [3]. GANs have been applied for generative modeling in

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a wide range of disciplines, including energy-related domains. For example, Zhang et al. developed SolarGAN [4], which generates synthetic annual solar irradiance time series datasets for urban building facades using fisheye images. Rizzato et al. [5] applied GANs to generate load scenarios for stress testing of electrical grids. Carreon et al. [6] employed GANs to generate realistic flame images resembling those from a combustor experiment, demonstrating that GANs could capture the statistical structure of the experimental data for a variety of flame configurations. Menon and Ranganathan [7] reviewed the state-of-the-art applications of popular DGMs, including GANs, for applications in biomaterials and energy materials. In a recent work, Nabila et al. [8] utilized a Wasserstein GAN variant for energy demand forecasting.

In the early 2020s, the undisputed dominance of GANs began to wane with the rise of other highly effective DGMs, especially DMs. DMs are inspired by non-equilibrium thermodynamics [9], and they use a Markov chain of diffusion steps to slowly add random noise to the data and then learn to reverse the diffusion process to construct desired data samples from the noise. A series of works have shown that DMs beat GANs with image generation quality and training stability, in tasks including image synthesis [10], topology optimization [11], talking-face generation [12], and image classification [13]. Although DMs typically have a longer training time than GANs due to the need for a large number of denoising steps during sample generation in the reverse diffusion process [14], some of the state-of-the-art text-to-image generation tools have been developed based on DMs, such as Dall-E [15], stable diffusion [16], Imagen [17] and Midjourney. Additionally, DMs are adaptable to tasks beyond simple image generation and have been utilized for anomaly detection [18] and time series imputation and forecasting [19]. DMs have proven to be application-agnostic and have been applied for diverse uses such as in the medical field for image-to-image generation [20] and resolution enhancement of diagnostic scans [21], fault diagnosis in mechanical systems [22], and nuclear physics [23].

Most applications of DMs have focused on text, image, or video data, and their applications for data augmentation of scientific datasets, particularly in the energy sector, have been relatively limited. For example, Zhang et al. [24] developed a physics-informed DM for generating synthetic net load data, addressing the challenges of data scarcity and privacy concerns. Jiang and co-authors [25] developed a multivariable DM for transportation energy demand prediction that can capture time-series tendencies while considering contextual information (e.g., COVID-19). Other examples can be found in inverse design of specialized materials for energy applications. Bastek and Kochmann [26] proved that video denoising DM trained on full-field data of periodic stochastic cellular structures can be used for inverse design of nonlinear mechanical metamaterials. Park et al. [27] developed a DM for the generation of novel crystalline porous materials.

Studies of deep generative learning in nuclear energy are relatively recent and limited, despite the need for AI-driven solutions in this safety-critical domain. Nuclear energy systems present unique challenges that make them ideal testbeds for advanced AI methodologies like DMs: stringent safety requirements, limited experimental data due to operational constraints and costs, and the necessity for physics-consistent predictions. These constraints necessitate DGM approaches that can augment data while preserving underlying physical relationships, a capability which is relevant across the broader energy sector. In a previous work [28], we compared the performances of GANs, normalizing flows, VAEs and conditional VAEs to generate synthetic void fraction data using a classical benchmark on boiling water reactor bundle test. It was found that conditional VAEs achieved the smallest errors in generating scientific data. In a follow-up study [29], we demonstrated that ML models trained using the augmented dataset can achieve much better predictive accuracy and reduce predictive uncertainties. In another recent study [30], we investigated the conditional VAE model for predicting critical heat flux (CHF) with uncertainty quantification (UQ) and domain generalization. The numerical results demonstrated that a conditional VAE performs better than conventional deep neural networks in predicting CHF and exhibits more reliable uncertainty estimates. A similar problem was studied in [8] using conditional GANs for constrained synthetic data generation. Very recently, Liu et al. [31] employed a DM for the development of an unsupervised anomaly detection method for nuclear power plants, and demonstrated that the DM outperformed autoencoders, VAEs, and GANs in detection accuracy. In the work by Barra et al. [32], the authors presented a workflow for the inverse design of molten salts with targeted density values. The authors used a dataset of critically evaluated molten salt densities to train a VAE coupled with a predictive deep neural network, which then can be used to generate new molten salt compositions with desired density values. This work can efficiently exploit molten salts' customizability and unlock advantages in energy production and energy storage.

The main objective of this work is to investigate the effectiveness of DM in generating physics-consistent data for CHF, which is one of the most important safety-related limiting quantities in nuclear energy. The CHF experimental dataset used for training is the public dataset that was used to develop the widely known 2006 Groeneveld CHF lookup table [33]. The dataset was curated and published by the U. S. Nuclear Regulatory Commission (NRC) [34] and is the largest known CHF dataset publicly available with measurements in vertical water-cooled uniformly heated tubes. The dataset consists of 24,579 samples, and it is well suited for data augmentation studies, as it is sufficiently large to train a DGM, but still significantly smaller than the typical sample sizes for image generation tasks. We determine the degree of conformity with physical correlations through a rigorous validation framework which compares steam quality of the experimental Groeneveld data to the synthetic dataset, testing the maintenance of known TH relationships between features of the dataset.

A second objective of this work is to develop DGMs for targeted data generation. One major limitation of the vanilla DM is that it can only produce synthetic samples randomly. However, for practical applications it is desirable to have the synthetic data generated at specific conditions, in this work, CHF values at specified thermal-hydraulic (TH) conditions. To this end, we also developed a conditional DM (CDM) that generates CHF samples at given TH conditions. Conditional DGMs have been studied across various fields, with a wide range of success. For example, conditional GANs (CGANs) [35] were developed shortly after GANs. CGANs were applied [36] to generate building energy consumption data, which supports numerous downstream applications such as retrofit analysis, smart grid integration and optimization, and load forecasting. In another work [37], a conditional GAN was employed for producing short-term time series electricity price scenarios. Similarly, conditional VAEs models have proven highly effective in our preliminary work on CHF [30]. Other applications of conditional VAEs in the energy sector include [38] for simulating tariff impact on electrical energy consumption profiles, [39] for wind power curve modeling, and [40] for photovoltaic power generation to study sensing anomalies. The development and applications of conditional DMs remain relatively limited compared to conditional GANs and VAEs, primarily because DMs have only gained popularity in the early 2020s. However, several energy-related studies exist in the literature. For example, [41] for short-term wind power scenario generation, [42] for synthetic energy meter data using power meters from various buildings and countries, and [43] for customized load profile synthesis for electricity customers. To the authors' knowledge, this work represents the first application of a DM and conditional DM in the nuclear energy domain.

The performances of the DM and CDM were evaluated based on the similarity between the synthetic samples and the real CHF samples from the training dataset. We compared the marginal distributions of both the TH features and the CHF output, as well as the pair-wise correlations between them. We applied various error metrics to further assess the performance of the DGMs. The results demonstrate that both

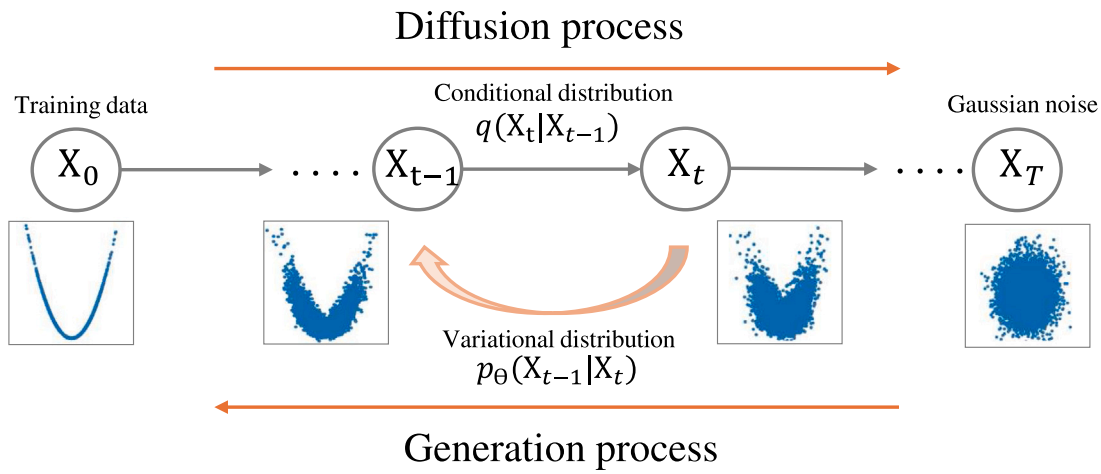


Fig. 1. Illustration of the diffusion and generation processes of a DM.

the DM and CDM models successfully generate CHF data by accurately learning the empirical marginal distributions and correlations between features and CHF output. To establish confidence in the generated samples, we performed UQ analysis by leveraging the inherent variability in the CDM generation process. During generation, the CDM creates new samples from random noise, which is removed step by step according to the given conditions. By varying the initial noise input each time, the model produces different outputs for the same set of conditions, allowing us to quantify the uncertainties in the outputs. The CDM achieved high predictive accuracy for CHF values generated under user-specified conditions.

The novelty and contributions of this work can be summarized as: (1) development of a DM for generative learning for a critical safety-related limiting quantity in nuclear energy, (2) extension of the DM to a CDM for targeted synthetic data generation, (3) a comprehensive performance evaluation of the DGM results using various quantitative metrics, including an assessment of physics-consistency, and (4) UQ analysis of the generative process to quantify uncertainty in the produced samples.

The remainder of this paper is organized as follows: Section 2 presents an overview of the DM and CDM methodologies. Section 3 establishes the application and scope of this study and describes the training and application of the DM and CDM for generative analysis. Section 4 discusses the results obtained using the DM, while Section 5 analyzes the CDM results, including the UQ analysis and physical-consistency evaluation. Finally, Section 6 summarizes the findings and concludes the paper.

2. Methodologies

2.1. Diffusion model

DMs draw inspiration from the principles of non-equilibrium thermodynamics [9]. The training process of these models consists of two distinct phases: the diffusion, or "noising" process, and the generation or "denoising" process, as illustrated in Fig. 1. During training, the model learns the underlying data distribution by successively adding Gaussian noise to the training data. The model then learns to reverse this noising process, aiming to accurately recover the original data [44]. After training, DMs are utilized to generate new samples simply by passing random noise through the learned denoising process.

Given a data point $\mathbf{x}_0$ sampled from a real data distribution $\mathbf{x}_0 \sim q(\mathbf{x})$, the forward diffusion process progressively adds small amounts of Gaussian noise to sample $\mathbf{x}_0$ over T steps (also referred to as time-steps), producing a sequence of noisy samples $\{\mathbf{x}_t\}_{t=1}^T$. To avoid confusion with the "time steps" in transient modeling and simulation problems, in this work we will refer to T as the number of "time-steps". This process follows a sequential transition kernel $q(\mathbf{x}_t|\mathbf{x}_{t-1})$, formulated as follows:

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t \mathbf{I}) \tag{1}$$

where $\{\beta_t\}_{t=1}^T \in (0,1)$ denotes the variance schedule, which controls the amount of noise added at each time-step and is also referred to as the step size. The original data sample $\mathbf{x}_0$ gradually loses its characteristics as the step t becomes larger. When $T \to \infty$, $\mathbf{x}_T$ converges to an isotropic Gaussian distribution. In practice, a very large T is typically unnecessary to achieve this approximation.

The diffusion process can be collectively defined through a series of transition kernels:

$$q(\mathbf{x}_{1:T}|\mathbf{x}_0) = \prod_{t=1}^{T} q(\mathbf{x}_t|\mathbf{x}_{t-1}). \tag{2}$$

We can recreate the true samples from a Gaussian noise input, $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, by reversing the forward diffusion process and sampling from $q(\mathbf{x}_{t-1}|\mathbf{x}_t)$. However, in practice $q(\mathbf{x}_{t-1}|\mathbf{x}_t)$ is intractable. Therefore, we choose to learn a model $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ to approximate these conditional probabilities in order to run the reverse diffusion process. The generation process entails removing noise at each step in the reverse order until the original training data is reconstructed. Starting with pure Gaussian noise $p(\mathbf{x}_T) = \mathcal{N}(\mathbf{x}_T; \mathbf{0}, \mathbf{I})$, the model learns the joint distribution $p_\theta(\mathbf{x_{0:T}})$:

$$p_\theta(\mathbf{x}_{0:T}) = p(\mathbf{x}_T) \prod_{t=1}^{T} p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) \tag{3}$$

where θ denotes the learnable parameters. By assuming that the conditional distributions are approximately Gaussian, we have

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}\left(\mathbf{x}_{t-1}; \boldsymbol{\mu}_\theta(\mathbf{x}_t, t), \boldsymbol{\Sigma}_\theta(\mathbf{x}_t, t)\right) \tag{4}$$

where the mean vector $\boldsymbol{\mu}_\theta(\mathbf{x}_t, t)$ and the covariance matrix $\boldsymbol{\Sigma}_\theta(\mathbf{x}_t, t)$ are parameterized by deep neural networks.

The training setup for a DM is very similar to that of a VAE, since $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ can be treated as a variational distribution for $q(\mathbf{x}_{t-1}|\mathbf{x}_t)$. We train the DM using the simplified loss function derived by Ho et al. [44]:

$$\mathcal{L}_t = \mathbb{E}_{t\sim[1,T],\mathbf{x}_0,\boldsymbol{\epsilon}_t}\left[\|\boldsymbol{\epsilon}_t - \boldsymbol{\epsilon}_\theta(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}_t, t)\|^2\right] \tag{5}$$

where $\boldsymbol{\epsilon}_t$ is the true noise added to the data, and $\boldsymbol{\epsilon}_\theta$ is the noise predicted by the model. The loss function computes the mean squared error between the true and predicted noise. All symbols used in this section are defined in Table 1.

Table 1
Definitions of mathematical symbols for DM.

Symbols	Meanings
$q(\mathbf{x})$	Real data distribution
$\mathbf{x}_0$	Data point sampled from real data distribution
$\mathbf{x}_t$	Noised latent variable at time-step t
$\boldsymbol{\mu}_\theta(\mathbf{x}_t, t)$	Mean function at time-step t
$\boldsymbol{\Sigma}_\theta(\mathbf{x}_t, t)$	Covariance function at time-step t
β_t	Variance schedule
α_t	$1 - \beta_t$
$\boldsymbol{\epsilon}_t$	Gaussian noise at time-step t
θ	Learnable parameters
$\mathcal{L}_t$	Divergence between inferred state at $\mathbf{x}_t$ and the true distribution at t
ϵ_θ	Function approximator to predict ϵ from $\mathbf{x}_t$

2.2. Conditional diffusion model

The vanilla DM can only generate data randomly and cannot target specific conditions. To address this limitation, we developed a CDM that is capable of targeted synthetic data generation at user-provided TH conditions. A CDM follows the same general framework as a DM, beginning with the successive addition of Gaussian noise. In a CDM, the denoising process is conditioned on additional data, guiding the model to generate data under user-specified conditions. After training, new data can be generated by passing a noise vector and a "condition vector" to the model.

By modifying the DM loss function from Eq. (5), the loss function used to train a CDM can be expressed as follows [10,45]:

$$\mathcal{L}_t = \mathbb{E}_{t\sim[1,T],\mathbf{x}_0,\boldsymbol{\epsilon}_t}\left[\|\boldsymbol{\epsilon}_t - \boldsymbol{\epsilon}_\theta(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}_t, t, \mathbf{c})\|^2\right]. \tag{6}$$

The loss function remains the mean squared error between true and predicted noise. In this case, $\boldsymbol{\epsilon}_\theta$ represents the model-predicted noise conditioned on both the given noisy input and the condition vector $\mathbf{c}$.

CDMs can be highly beneficial when a direct comparison between real and generated data is needed as they facilitate the direct calculation of errors and the estimation of uncertainties. During generation, the model receives a random vector and user-provided TH conditions from a held-out test dataset not seen in the training process. By varying the random vector while holding the conditions fixed, different generated values can be obtained. This variability assists in estimating the model's uncertainty, thereby establishing the user's confidence in the generated values.

3. Problem definition

This section presents the problem definition and the CHF dataset. We will first describe the CHF phenomena and explain its importance for nuclear energy, followed by an overview of the CHF training dataset in Section 3.1. Sections 3.2 and 3.3 detail the training procedure of the DM and CDM models, respectively.

3.1. The CHF dataset

This study explores the capabilities of DMs and CDMs in generating realistic CHF data. CHF is one of the most important safety-related limiting quantities in nuclear energy. In the study of heat transfer in nuclear reactors, CHF consists of two distinct physical phenomena: departure from nucleate boiling, which occurs mainly in pressurized water reactors with low quality flows, and dryout, which occurs in high quality conditions typically found in boiling water reactors. In nuclear reactor operations, exceeding the CHF limit must be avoided as it can lead to fuel rod failure. Exceeding CHF indicates that the heated fuel rod surface has reached a point where it can no longer efficiently transfer heat to the surrounding fluid. Therefore, data augmentation to expand the CHF experimental database is essential for the design and safety analysis of advanced water-cooled reactors.

The DM and CDM were trained on the largest known publicly available CHF experimental dataset, curated and published by the US NRC [34]. The dataset comprises 24,579 CHF measurement data points from vertical uniformly-heated water-cooled tubes, compiled from 59 separate experimental campaigns performed over a span of 60 years. These experiments utilized various CHF identification methods, such as visual identification, physical burnout, changes in the test section resistances, and thermocouple measurements. The parameter space covered by this CHF database is substantial as it spans a wide range of commercial nuclear reactor operational conditions. It is worth noting that recently the same dataset has been used for an international benchmark on "Artificial Intelligence and Machine Learning for Scientific Computing in Nuclear Engineering" [46], organized by the Organization for Economic Co-operation and Development (OECD) Nuclear Energy Agency (NEA). The benchmark tasks included feature analysis and regression-based ML model development for CHF prediction. This work on synthetic CHF dataset generation can significantly expand the size and diversity of the available CHF data, thereby improving the performances of such predictive ML models.

The CHF training dataset includes various TH parameters (also referred to as the features), including initial and boundary conditions such as pressure (P), mass flux (G), and inlet temperature (T_{in}), and geometrical parameters like test section diameter (D) and heated length (L). The dataset also includes derived parameters from measurements and water properties, including outlet equilibrium quality (x) and inlet subcooling (h_{sub}). The distributions of these parameters, along with their pair-wise correlations, are shown in Fig. 2. In this work, we have selected five TH conditions as the input features P, G, D, L, and x, along with CHF as the target output. This selection is consistent with the recommendations from the OECD/NEA CHF benchmark summary [47], as predictive ML models with these input features achieve optimal performance.

3.2. Training of DM

DGM training is essentially unsupervised, where the inputs (P, G, D, L, and x) and outputs (the CHF values corresponding to the five-dimensional TH conditions) of a training sample are represented as a single six-dimensional vector, without explicitly learning the functional mapping from inputs to outputs. The DGM learns the joint distribution of all six parameters together. Generated synthetic samples maintain the same dimensionality, preserving the CHF-TH-parameter relationships. During generation with a DM, random noise is used to produce samples consisting of five TH parameters and the corresponding CHF value. The major limitation of a vanilla DM is that direct evaluation of accuracy of the generated CHF values by using a held-out test dataset is impossible, because the synthetic samples are randomly generated. Therefore, we evaluate the vanilla DM performance by assessing the following:

1. The ability to capture the distributions of each TH parameter and the CHF output. This is done by comparing the marginal distributions of the true and generated samples.
2. The ability to capture the correlations between parameters in the dataset, evaluated by comparing the Pearson correlation coefficient (PCC) and Spearman rank correlation coefficient (SRCC) of the measurement and generated datasets.
3. The generation of physically meaningful values for all parameters.
4. The overall distributional similarity between the generated data and the training data, assessed by comparing the full joint empirical cumulative distribution functions (ECDFs) of the real and synthetic datasets.

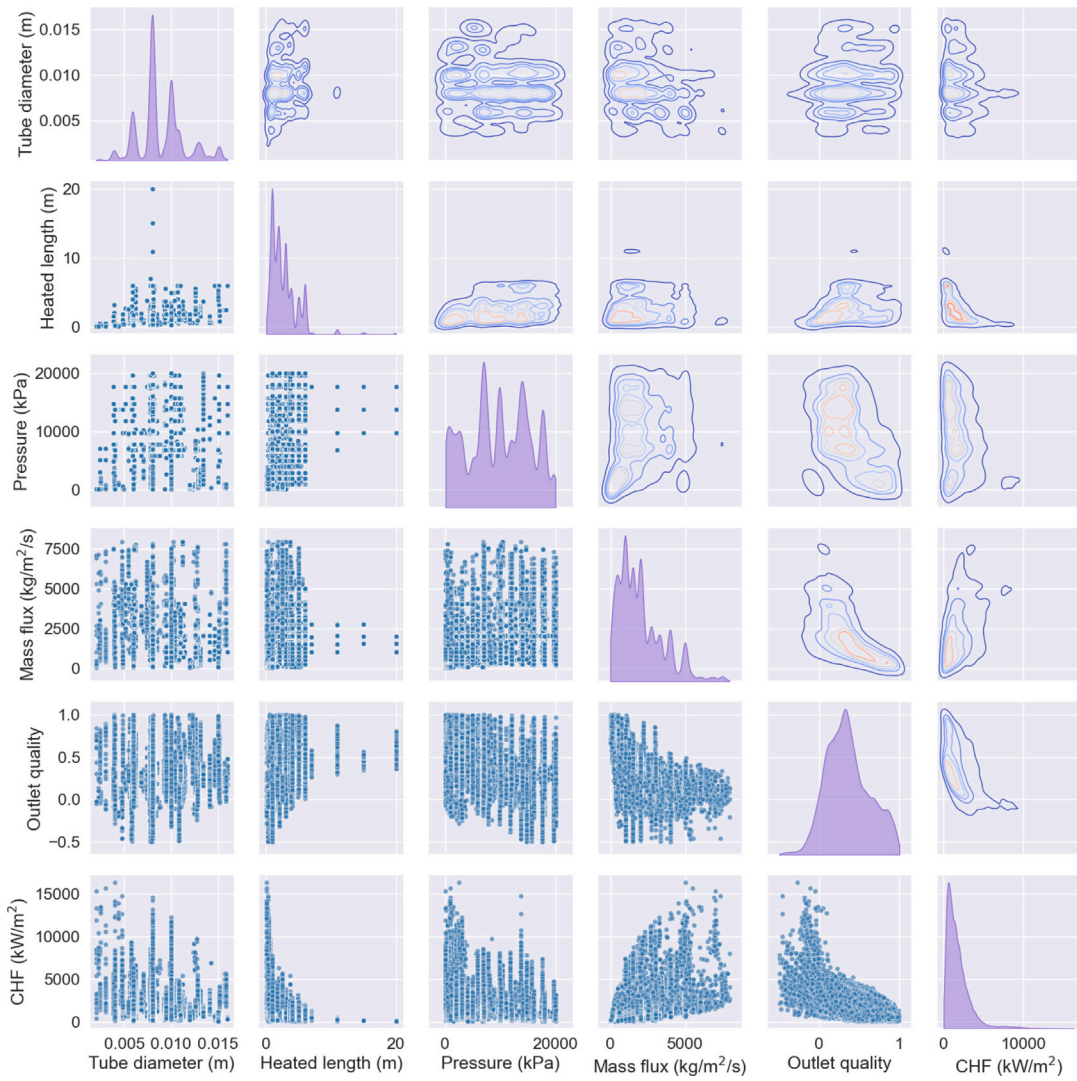


Fig. 2. The distributions and correlations of the TH parameters and CHF values in the NRC CHF dataset.

These metrics can evaluate whether the model has learned the underlying distribution of the training data, and whether the synthetic samples can be considered as drawn from the original distribution.

The DM was trained using the simplified loss function utilized by Ho et al. in the fundamental paper for denoising diffusion probabilistic models [44] as shown in Eq. (5). To stabilize training, the exponential moving average (EMA) technique was used. Rather than directly updating the model's parameters, a copy of the previous parameters is maintained and updated as a weighted mean between the current and previous values. Incorporating the EMA technique improves training stability by reducing the impact of individual parameter updates. EMA is calculated as a weighted combination of the current parameter values and their previous values:

$$\text{EMA} = (1-\mu)\times \text{parameter}_{\text{current}} + \mu \times \text{parameter}_{\text{previous}} \tag{7}$$

where μ is the smoothing constant, with a value between 0 and 1. A lower value gives more weight to the current parameter, while a higher value gives more weight to the previous parameter. During the training of the DM, we used a value of 0.9, as suggested in [44].

The model used $T = 100$ time-steps for the diffusion process. Gaussian noise was progressively added using a sigmoid variance schedule (β), with noise values smoothly transitioning between 1×10^{-5} and 1×10^{-2}. Smaller noise values were added at the beginning of the diffusion process, while larger noise values were added in the later time-steps, following the shape of a sigmoid curve. The DM was trained for 1200 epochs with a batch size of 64 and a learning rate of 1×10^{-3}.

3.3. Training of CDM

We trained a CDM to generate CHF values under five user-specified TH conditions: P, G, D, L, and x. The dataset was split into 80% for training, 10% for hyperparameter tuning, and 10% for blind testing. By inputting TH condition values from the held-out test dataset, the CDM generates CHF values corresponding to these specified TH conditions. The model is then evaluated through direct comparison with the true CHF values in the test dataset not seen by the model during the training or hyperparameter tuning steps. Fig. 3 illustrates the workflow for CDM training.

UQ analysis is performed by leveraging the inherent stochasticity of the CDM generation process. Different samples of random noise fed to the isotropic Gaussian distribution, combined with the same TH conditions, will produce varying outputs through the generation process. To quantify uncertainty in the generated data, we repeat the generation process 500 times, providing a different random vector each time while keeping the conditions fixed. We then compute the mean of the 500 generated samples (μ_{samples}) for each condition vector in the test dataset, along with the standard deviation (σ_{samples}), which provides an estimate of the uncertainty in the generated data.

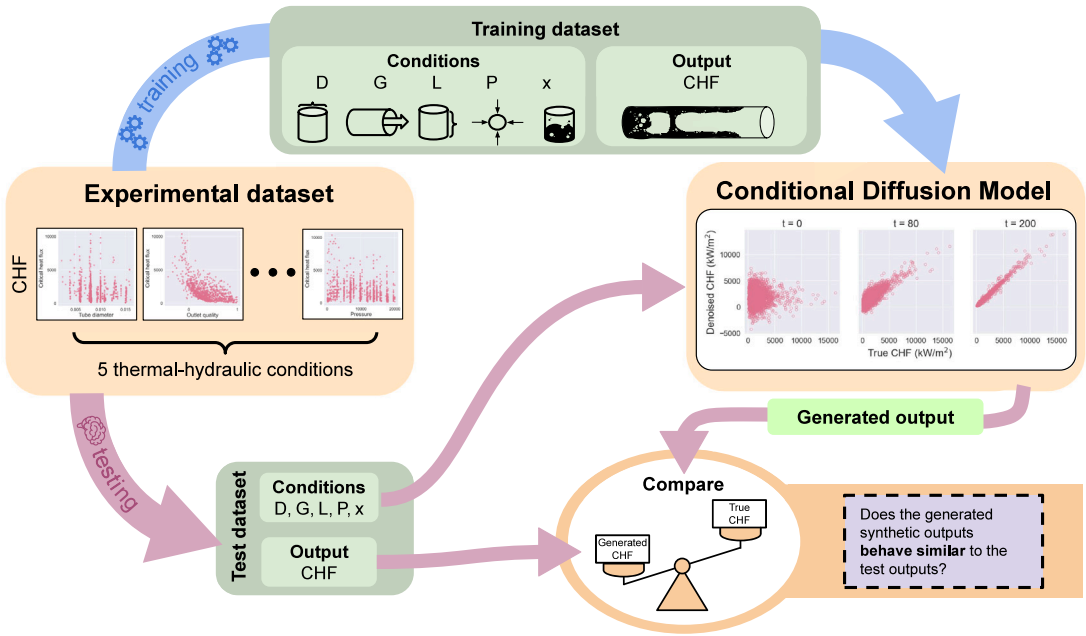


Fig. 3. Illustration of the CDM training process.

Table 2
Hyperparameter configurations for the DM and CDM models.

Hyperparameter	CDM	DM
Variance schedule	Sigmoid	Sigmoid
Noise range	10^{-5} to 10^{-2}	10^{-5} to 10^{-1}
Number of time-steps	200	100
Number of hidden layers	6	2
Hidden layer size	[400, 700, 900, 700, 400]	[260, 128]
Batch size	300	64

The CDM was trained using the loss function presented in Eq. (6). To stabilize the training process, EMA was also applied, as discussed in Section 3.2. For this model, the EMA value was set to 0.9. The model architecture consisted of six hidden layers, and a learning rate of 10^{-4} was used. The model was trained for 7500 epochs with a batch size of 300. Gaussian noise was added in each time-step using a sigmoid variance schedule (β) ranging from 10^{-5} to 10^{-2} over $T = 200$ time-steps. Hyperparameter tuning was performed for the variance schedule, noise range, number of time-steps, and model-specific hyperparameters (e.g., number of hidden layers, batch size). The resulting hyperparameters are presented in Table 2 for transparency.

4. Results using DM

A DM was trained to augment the CHF dataset by taking random Gaussian noise and generating synthetic samples, each consisting of values for the five TH parameters and the corresponding CHF. In this work, 10,000 synthetic samples were generated for the assessment of the DM. While the DM can generate an arbitrary number of new samples at an increased cost, one cannot control what samples will be generated. That is, the vanilla DM will generate new samples randomly, rather than at specific TH conditions provided by the user. As a result, we cannot evaluate its generative accuracy by a direct comparison with real CHF values. Instead, we compared the collective statistical behavior of the generated samples with the real experimental samples.

Fig. 4 compares the kernel density estimates of the five TH parameters and CHF, for experimental and DM-generated datasets. The distributions of some TH parameters were found to be smoother in the synthetic data. For instance, in the case of tube diameter (D), the measurement data is derived from experiments with discrete values for D, leaving gaps for values where no experiments were conducted. The generated data contains D values that do not exist in the measurement data while preserving the overall distribution shape. This observation also applies to the heated length, pressure, and mass flux. The outlet quality and the CHF values exhibit similar smooth distributions between the two datasets, due to the values in the training dataset representing a relatively continuous coverage of the parameter domains.

To assess whether the model successfully learned the correlations between CHF and the TH parameters in the real dataset, and can subsequently generate values while preserving these correlations, Fig. 5 shows a visual comparison of the CHF-TH-parameter pairwise correlations between the real data and the data generated by the DM. The generated data notably preserves the correlations between CHF and the TH parameters when compared to the real data, while also producing new samples not present in the training dataset.

Beyond the visual comparison shown in Fig. 5, we also calculated the pairwise PCCs and SRCCs for all five TH parameters and CHF in the training data, and compared each correlation metric with the corresponding values from the generated data. The PCC captures linear relationships, while the SRCC assesses monotonic relationships, which can be either linear or non-linear. Both metrics were used to evaluate whether the model successfully learned the correlations between the six variables (5 TH parameters and CHF). The comparisons between the PCC values and the SRCC values are shown in Figs. 6 and 7, respectively. The values indicate that the DM was able to capture the correlations between the parameters, as the PCCs and SRCCs calculated from the training data agree well with those from the generated data. Minor deviations are observed in some pairs of variables; however, these deviations are negligible.

The DM was also evaluated by testing the overall distributional similarity between the generated data and the real data by comparing the ECDFs of the six variables (five TH parameters + CHF). A comparison of the ECDFs between the real data and DM-generated data in Fig. 8 shows a strong agreement between the ECDFs.

In addition to a visual inspection of the six marginal ECDFs in Fig. 8, we utilized the genai-evaluation library for a quantitative comparison of the joint ECDF of all the six variables. This library was designed specifically for evaluating the quality of tabular synthetic data. First, we calculated the multivariate ECDFs of both datasets. Afterward, we compared the similarity between the real and generated data by

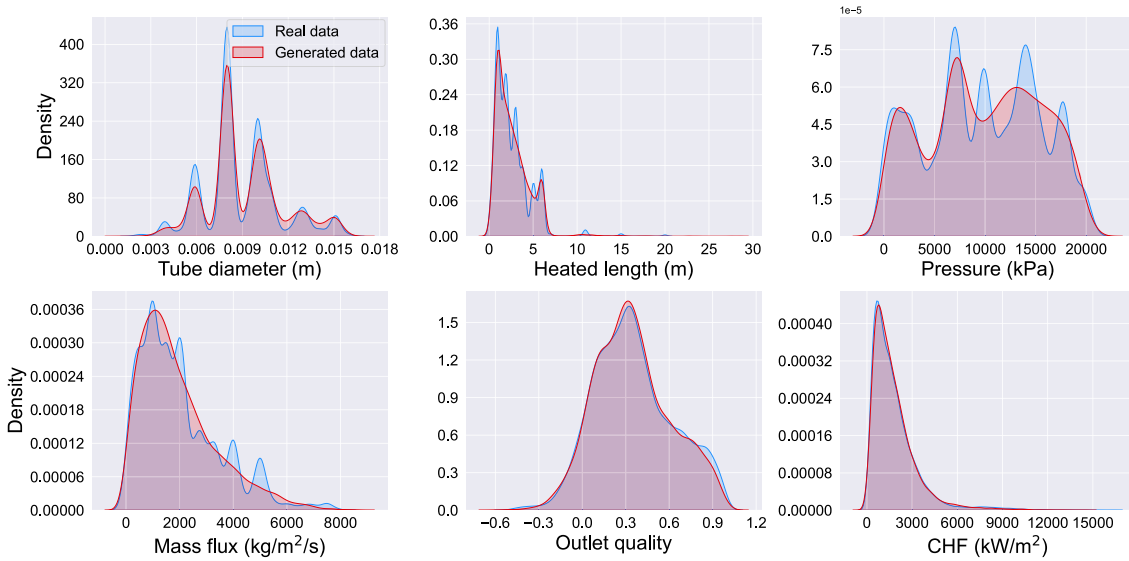


Fig. 4. Kernel density estimates of the TH parameters and CHF distributions for real and DM-generated data.

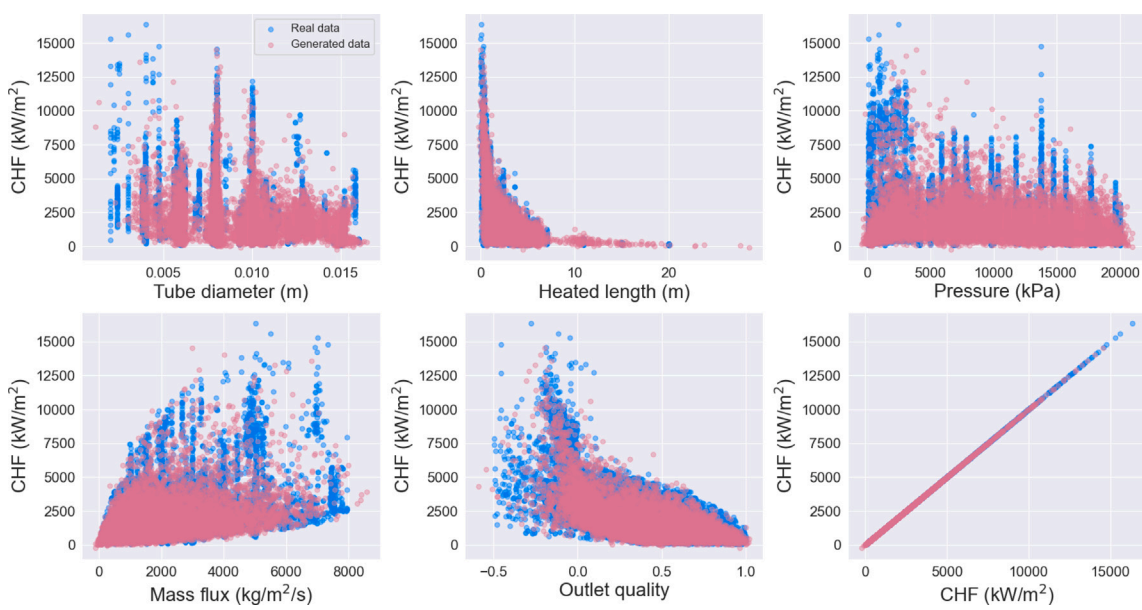


Fig. 5. Scatterplots of the CHF-TH-parameter pairwise correlations comparing the experimental and DM-generated data.

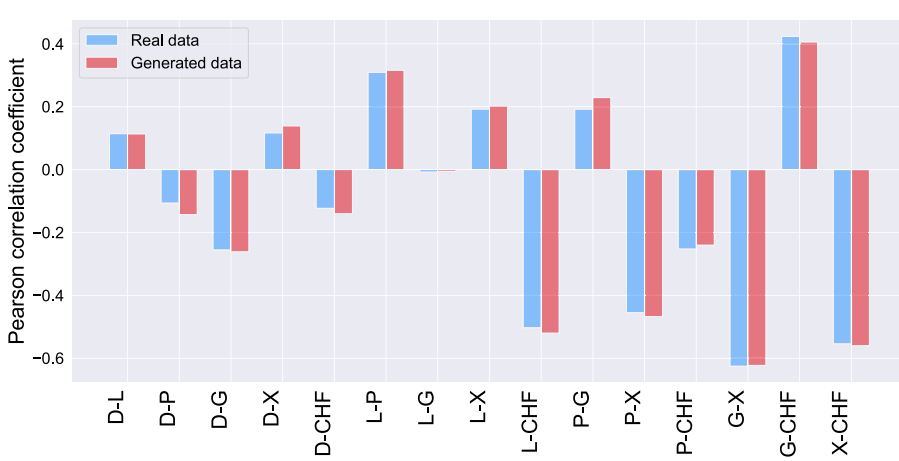


Fig. 6. Comparisons of PCCs of the CHF-TH-parameters between the real and DM-generated data.

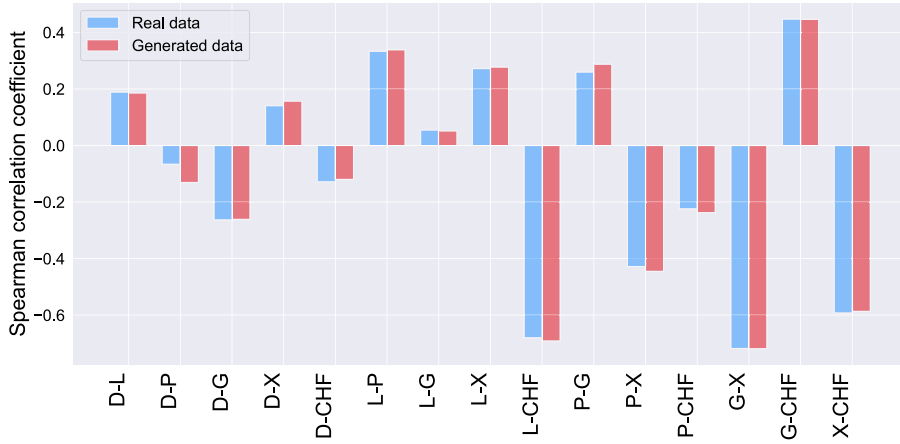


Fig. 7. Comparisons of SRCCs of the CHF-TH-parameters between the real and DM-generated data.

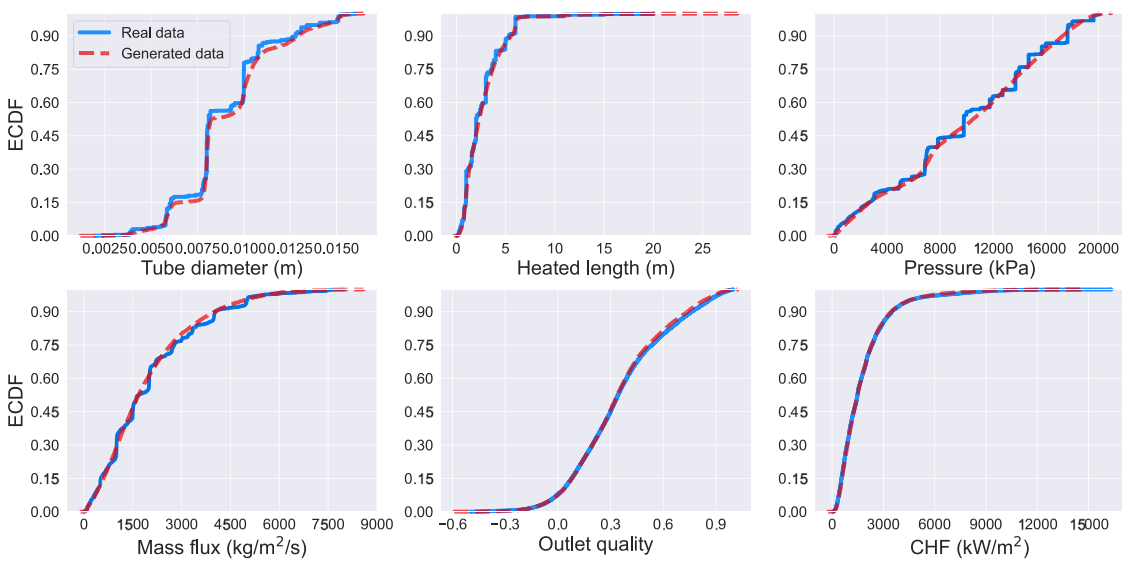


Fig. 8. Comparisons of the marginal ECDFs of the TH parameters and CHF for real and DM-generated data.

Table 3
Number of negative values of the TH parameters and CHF in the DM-generated data.

Parameters	Tube diameter	Heated length	Pressure	Mass flux	CHF
Number of negative values	0	6	27	9	1
Percentage (%)	0	0.06	0.27	0.09	0.01

computing the Kolmogorov–Smirnov (KS) distance between the two ECDFs. This metric produces a value between 0 and 1, where values close to 0 indicate that the two distributions are similar, while a value of 1 corresponds to distributions that are completely dissimilar. The KS distance between the two joint ECDFs is 0.1265, demonstrating strong similarity between the joint ECDF of the generated data and that of the training data. The KS distance confirms that the DM successfully learned the distribution of the training data and generated realistic synthetic samples.

Finally, we evaluated the model's ability to generate physically meaningful values for all parameters. The model produced a very small number of negative values for the generated parameters, as shown in Table 3, where the percentage of negative values for each parameter is minimal. The total percentage of unphysical samples generated is approximately 0.43%, indicating that the model successfully generated physically meaningful values for all parameters.

5. Results using CDM

5.1. Results of CHF generation

In this subsection, we present the results of the CDM in generating CHF values. The major difference from the vanilla DM is that the CDM can generated targeted CHF values at user-provided TH conditions. In this work, 10% of the real CHF dataset was reserved for testing. Evaluating the trained CDM at the same TH conditions with the test dataset will generate CHF values corresponding to these TH conditions. This enables us to compare the real and synthetic CHF values directly to evaluate the accuracy of the generated CHF samples, which was not possible for the vanilla DM in Section 4. Therefore, in this subsection, we will present some extra results using quantitative metrics.

Fig. 9 presents a visualization of the denoising process, showing the generation of CHF data under the TH conditions from the testing dataset. It illustrates how the model gradually removes noise over 200

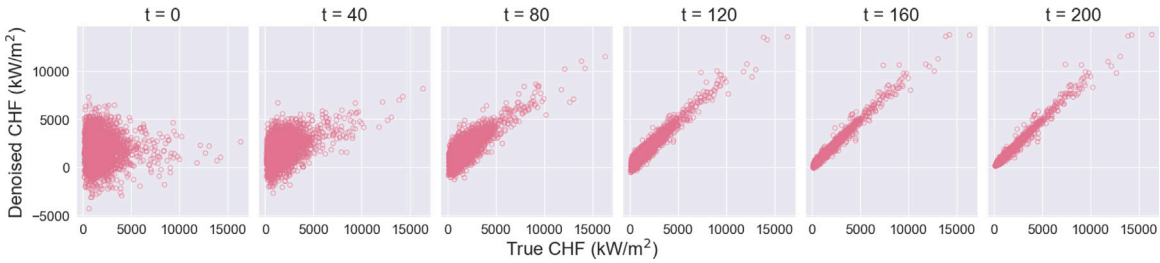
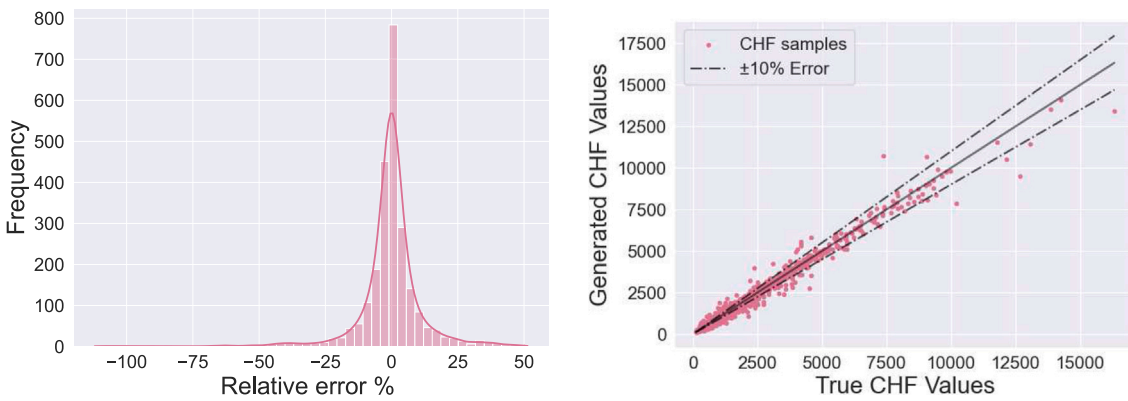


Fig. 9. Illustration of the denoising process of the CDM. Generation starts with pure random noise at $t = 0$ and gradually removes noise to generate realistic CHF values at $t = 200$.



(a) Relative error distribution (b) True vs. generated CHF values

Fig. 10. Performance assessment of CDM based on a direct comparison of the true vs. CDM-generated CHF values.

time-steps, starting from pure Gaussian noise at $t = 0$ and ending with the generated CHF values at $t = 200$. The figure displays the denoised samples every 40 time-steps against the true CHF values on the x-axis. Note that the noise and generated CHF values are shown after being rescaled, as the data was standardized using a standard scaler before being fed into the model.

The model's performance was evaluated by directly comparing the generated CHF values with the true CHF values from the held-out testing dataset. The relative error between the generated and true CHF values was calculated, and its distribution is presented in Fig. 10(a). The majority of errors in the generated samples fall within ±25%, with most testing points exhibiting small error values centered around the mean relative error of 6.86%. Additionally, less than 5.89% of the testing data show errors exceeding ±25%.

Fig. 10(b) presents a parity plot of the generated and true CHF values, including the ±10% error bounds. The generated CHF values show strong agreement with the true CHF values. To better interpret the results, several statistical metrics on the absolute relative errors, are presented in Table 4. These metrics include the mean, maximum, and standard deviation of the absolute relative errors, as well as the fraction of testing points with an absolute relative error exceeding 10%. The results show that the CDM accurately generated CHF values, and that while the maximum error was significantly higher than the mean, it occurred at very small CHF values, where small deviations from the true value can lead to a high relative error. Notably, only 5.89% of the data had errors greater than 25%, while 19.32% had errors greater than 10%. Furthermore, the R^2 value of approximately 0.98 indicates a strong correlation between the true and generated values.

Fig. 11 presents the pairwise correlations between CHF and the TH parameters for both the CDM-generated and real data. The figure demonstrates a strong alignment between the real and generated CHF values. To avoid redundancy, equivalent results to those presented in Section 4 will not be shown here. Overall, these results prove that the CDM performs well in generating CHF values based on the testing dataset.

Table 4
Statistical metrics for the absolute relative errors between the real and CDM-generated CHF values for the testing dataset.

Statistical metrics	Base case (Section 5.1)	With UQ (Section 5.2)
μ_{error}	6.86%	5.89%
$\text{Max}_{\text{error}}$	111.84%	113.55%
$\text{Std}_{\text{error}}$	10.64%	8.66%
$F_{\text{error}} > 10\%$	19.32%	18.18%

The results shown throughout this section indicate that the increased complexity of CDMs is worthwhile based on the observed improvements to the quality of the synthetic data. The Wasserstein distance of the synthetic CHF data compared to the experimental data from the Groeneveld dataset can be calculated as a quantitative metric to summarize the increased statistical fidelity. The CVAE-generated data from our previous DGM study was used as a baseline comparison for a simpler DGM baseline [30]. The CDM-generated CHF data improved the Wasserstein distance to 27.35 from the CVAE distance of 56.87, an approximately 52% improvement in similarity to the original data.

5.2. UQ analysis

Having established the CDM's generative performance, we next evaluate its uncertainty behavior. UQ is essential for assessing the reliability and establishing confidence in the generated data. UQ of CDM was performed as discussed in Section 3.3. Using the trained CDM, 500 samples were generated for each data point in the testing dataset. In other words, for each TH condition defined by P, G, D, L, and x, the generation process of the CDM was evaluated 500 times to get an ensemble of 500 CHF values, which serve as Monte Carlo samples for

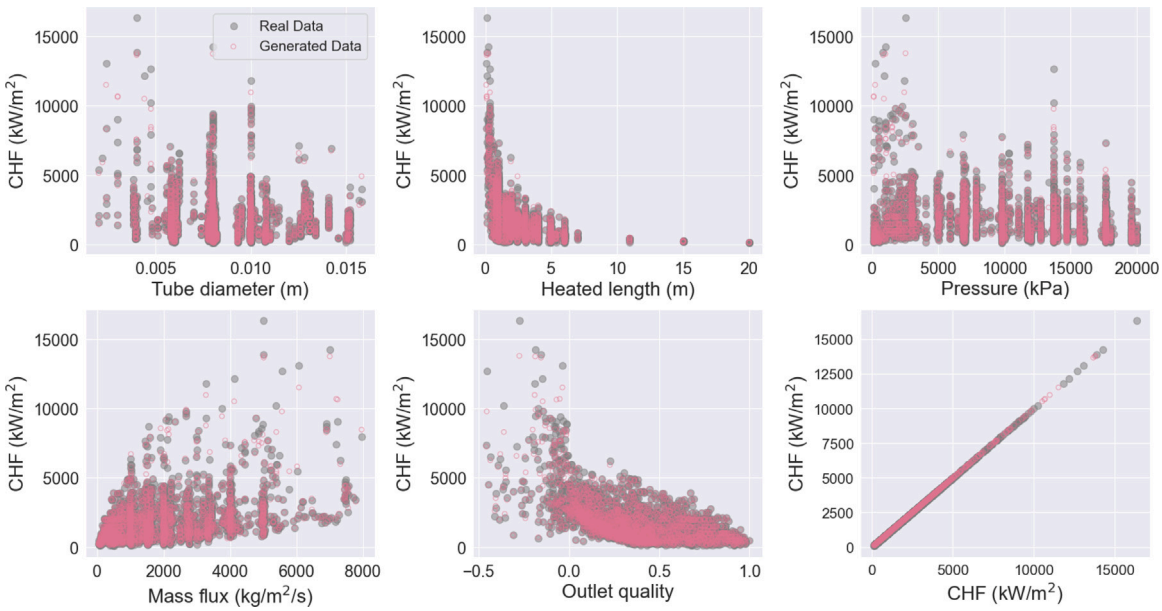


Fig. 11. Scatterplots of the CHF-TH-parameter pairwise correlations comparing the real and CDM-generated data.

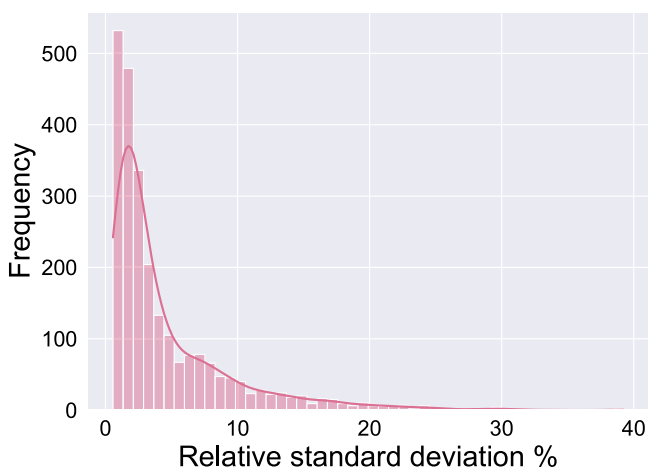


Fig. 12. Distribution of the relative standard deviations for UQ analysis of the CDM-generated CHF values based on the testing dataset.

the CHF at the chosen TH condition. The randomness comes from the generation of Gaussian noise at $t = 0$ in the generation step. The mean (μ_{samples}) and standard deviation (σ_{samples}) of these samples were then computed, and the relative standard deviation was calculated using Eq. (8) to allow for better comparison across different scales of outputs.

$$\text{Relative Std (\%)} = \frac{\sigma_{\text{samples}}}{\mu_{\text{samples}}} \times 100\% \tag{8}$$

Fig. 12 presents the distribution of relative standard deviation values across the testing dataset. The majority of relative standard deviation values are concentrated around the mean value of 4.40%, with most values remaining below 10%. This indicates that the CDM produces relatively stable predictions with low variability. Only a few data points exhibit significantly higher relative standard deviation values, with a maximum of 44.47%, suggesting the existence of some instances where the model has higher variability in the generated values. Importantly, many CHF values in the testing dataset are very small, as shown in Fig. 11. For these cases, a small standard deviation value can result in a large relative standard deviation.

Table 4 shows the same error metrics used in Section 5.1, with the errors calculated by comparing μ_{samples} with the true CHF value. The results from Table 4 also show improvement in the μ_{error} values dropping from 6.86% to 5.89%, when compared to the base case (without UQ), indicating slightly increased accuracy in CDM-generated values. This held true for the other metrics with the exception of the maximum error, which slightly increased.

To evaluate model confidence, the prediction interval coverage (PIC) is plotted in Fig. 13. PIC estimates the number of true CHF values that lie in confidence bounds based on the synthetic CHF data. For a perfectly calibrated model, the number of points in the synthetic data confidence bound will be the same for the empirically observed CHF data. The CDM confidence bounds closely align with the experimental data, but the CDM consistently shows both a slight overestimation of the coverage and a conservative tendency to avoid predicting outliers.

5.3. Validation of physical consistency

To enable downstream ML-based predictive tasks using the synthetic data, a validation of the physical consistency is essential. The assessments performed in Sections 4 and 5.1 focused primarily on comparison of the statistical behavior of the real and generated datasets. Although there is a sizable literature on physics-informed ML (sometimes referred to as "physics-aware", "physics-constrained", "physics-guided", etc.), the methodology for embedding physics information in the training process is problem-dependent, as is the assessment of physical consistency. In this section, we propose an approach to validate the physical consistency of CDM-generated data using thermodynamic relationships from the public CHF dataset, as well as the physical dependence among the TH features.

We leverage theoretical and empirical thermodynamic models underlying the CHF phenomena that describe the inter-parameter physical dependencies. The outlet equilibrium quality (x) can be obtained via the outlet enthalpy using an energy balance equation [48], as well as by using steam tables to derive the saturated liquid and vapor enthalpies. In this work the XSteam library was used when the steam tables were needed. Eq. (9) presents the process to obtain the calculated outlet

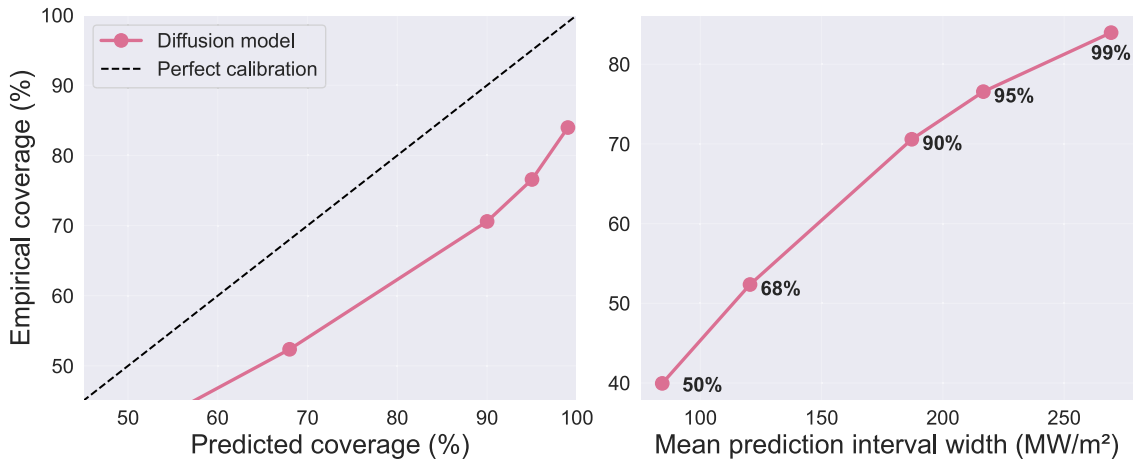


Fig. 13. Prediction coverage study of the CDM-generated CHF values compared to the empirical spread of the experimental data.

equilibrium quality (x) using other TH parameters and the CHF.

$$\begin{aligned} h_f &= h_{\mathrm{L}}(P) \\ h_{fg} &= h_{\mathrm{V}}(P) - h_f \\ h_{\mathrm{in}} &= h_f - h_{\mathrm{sub}} \\ \Delta h &= \frac{4\, q_{\mathrm{CHF}}\, L}{G\, D} \\ h_{\mathrm{out}} &= h_{\mathrm{in}} + \Delta h \\ x_{\mathrm{calc}} &= \frac{h_{\mathrm{out}} - h_f}{h_{fg}} \end{aligned} \tag{9}$$

In brief, to assess the physical consistency of the CDM-generated samples, we compare the outlet equilibrium quality (x) from the following three sources:

- "Measured": x is available from the public CHF dataset. It has been used as a TH condition for the DM and CDM models in Sections Section 4, 5.1, and 5.2. Note that x is not directly measured but derived based on CHF and other TH variable measurements and water properties, as mentioned in Section 3.1. It is referred to as "measured" here because it is available from the public CHF dataset, and to avoid confusion with "calculated" x below.
- "Calculated": x can be computed using other TH features together with the CHF values in the measurement data using Eq. (9). Note that there will be some discrepancies between the "Measured" and "Calculated" values for x due to two major reasons. Firstly, transcription errors have been common in CHF experimental dataset caused by either human mistakes or technological limitations, considering the long history of some of the CHF datasets. Secondly, many historical CHF measurements used older thermodynamic properties of water to derive the outlet equilibrium quality (x) and inlet subcooling (h_{sub}), for example, based on previous versions of the International Association for the Properties of Water and Steam (IAPWS) libraries. As a result, the newly calculated values for x will be different from those in the public CHF dataset.
- "Generated": x can be calculated using generated data from a CDM conditioned on D, G, P, L, and h_{sub} using Eq. (9). Note that this is a new CDM that is different from the CDM used in Sections Section 5.1, and 5.2, which was conditioned on D, G, P, L, and x.

To validate the physical consistency, we compared the absolute errors in x by comparing "measured - generated" and "calculated - generated", as shown in Fig. 14. This approach examines whether the consistency in x can be maintained through the combined calculation with Eq. (9) and generation by the CDM. Note that the CDM has been changed to be conditioned on a new TH parameter h_{sub} rather than x to enable x calculation from the generated h_{sub} values. Our earlier analyses used x instead of h_{sub} to be consistent with the recommendations from the OECD/NEA CHF benchmark summary [47].

Table 5
Comparison of statistical metrics of the absolute errors in the physical consistency evaluation.

Statistical metrics	Measured - generated	Calculated - generated
Mean	0.0197	0.0190
STD	0.0292	0.0288
Min	7.0e−9	6.0e−6
First quartile	0.0045	0.0043
Median	0.0114	0.0107
Third quartile	0.0242	0.0229
Max	0.5823	0.5834

A physically consistent DGM should exhibit comparable performance against measured and calculated data. The absolute error was selected as the metric on which to compare the discrepancies of the generated data, as a small portion of the x values are near zero, which biases performance metrics towards over-representation of outliers. Table 5 summarizes the results, showing nearly identical performance of the generated qualities. The distributions of the absolute errors are visualized in Fig. 15, which shows that the error distributions are very similar, and most absolute error values are small in magnitude.

6. Summaries and conclusions

Most applications of diffusion models (DMs) have focused on text, image, or video data, and their applications for data augmentation of scientific datasets, particularly in the energy sector, have been relatively limited. No prior work exists on DMs and conditional DMs (CDMs) for overcoming data scarcity in the nuclear energy domain. The main objective of this work is to investigate the effectiveness of DMs in generating physics-consistent data for critical heat flux (CHF), which is one of the most important safety-related limiting quantities in nuclear energy. By leveraging a public dataset on CHF that covers a wide range of commercial nuclear reactor operational conditions, we developed a DM that can generate an arbitrary amount of synthetic samples for augmenting the CHF dataset. Statistical analysis of the DM results showed that the DM effectively learned the distribution of the training data and generated realistic samples. Additionally, the model was able to capture the correlations between the parameters in the dataset, producing a negligible number of unphysical samples.

Another goal of this work is to develop deep generative models capable of targeted data generation. Since a vanilla DM can only generate samples randomly, we also developed a CDM that can generate

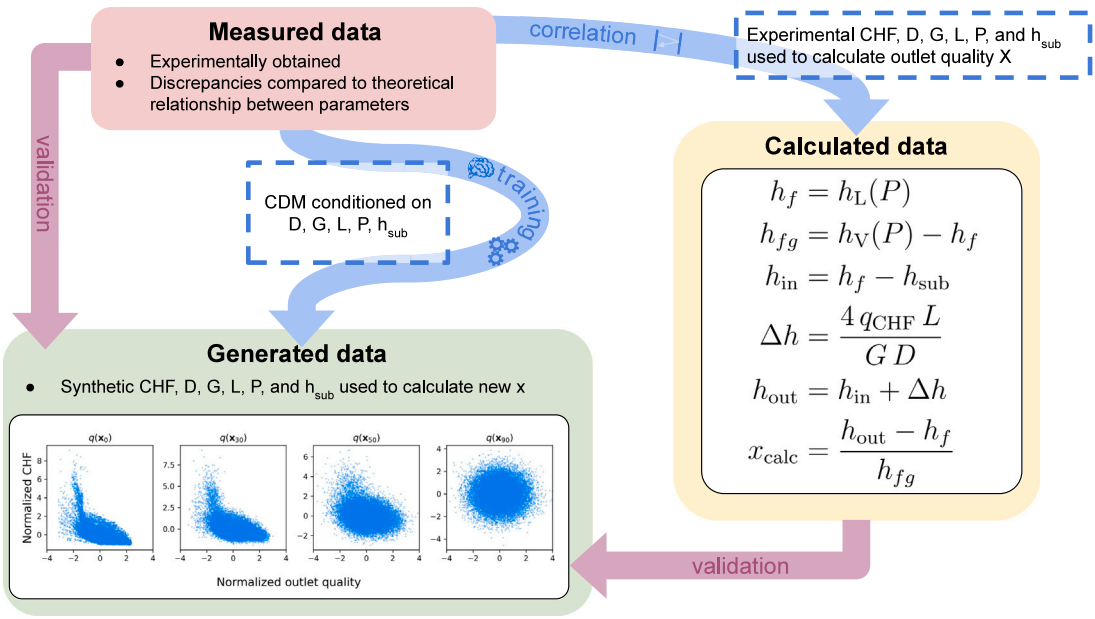


Fig. 14. Workflow for physical consistency assessment of the outlet equilibrium quality (x).

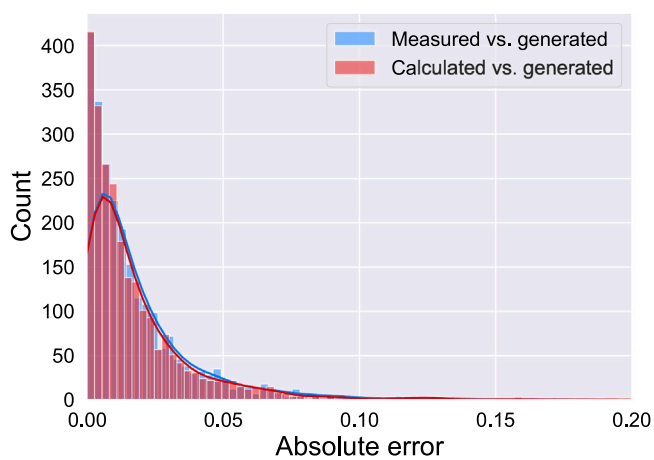


Fig. 15. Comparison of absolute error distributions of the generated outlet equilibrium quality with measured and calculated values.

targeted CHF data at user-specified thermal-hydraulics conditions. The performance of the CDM was evaluated through direct comparison with the CHF values in the held-out testing dataset. The results showed that the CDM successfully generated CHF data with small relative error values.

Finally, we developed a method to validate the physical consistency of the generated data using the available information from the public CHF dataset, as well as the physical dependence among the thermal-hydraulics features. We used theoretical and empirical mathematical models underlying the CHF phenomena that describe the physical dependence among the thermal-hydraulics parameters to prove that the CDM-generated values for the outlet equilibrium quality can maintain a good physical-consistency with CHF and the other thermal-hydraulics parameters.

Although the DM has proven effective in synthetic data generation, it must be noted that the DM-generated samples can only cover the gaps in the existing experimental dataset to a relatively small extent. The synthetic samples still tend to fall within the same ranges as the training dataset. The CDM is preferred because it is capable of targeted data generation. A limitation of this methodology is the risk of hallucination during extrapolation outside of the training regimes and geometric ranges of the experimental data. The further a data point is from the convex hull of the original dataset, the more guessing the model will have to do to represent the extrapolated correlations between features. Future work is still needed for data augmentation in extrapolated domains. A potential solution is the integration of a physics-informed DM framework in the training process, where relationships outside of training domains are inferrable. Our future plan also includes combining deep generative modeling and transfer learning to overcome data scarcity in nuclear energy. Overall CDMs pose a unique opportunity to address data scarcity for AI use in energy, and opens the door to numerous advanced machine learning applications within the field.

CRediT authorship contribution statement

Farah Alsafadi: Writing – original draft, Validation, Methodology, Formal analysis. **Alexandra Akins:** Writing – original draft, Validation, Formal analysis. **Xu Wu:** Writing – review & editing, Validation, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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